

Program Analysis 2025-26

Lecture #18

Control Flow Analysis



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Exercises

Exercises

Consider and add labels to the following expressions
Guess analysis results and verify them

$$e \triangleq ((\text{fn } x \Rightarrow (x + 1)) (3))$$

$$e \triangleq ((\text{fn } f \Rightarrow ((f\ 76) + (f\ 77))) (\text{fn } a \Rightarrow a))$$

$$e \triangleq \text{let } f = \text{fn } x \Rightarrow x + 1 \\ \text{in let } g = \text{fn } y \Rightarrow y + 2 \\ \text{in let } h = \text{fn } z \Rightarrow z + 3 \\ \text{in } (f\ g) + (f\ h)$$

Exercise 1

$$[app] (\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2} \\ (\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) : \\ (\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge \\ \hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$$

Consider and add labels to the following expression
Guess an analysis result and verify it

$$e \triangleq ((\text{fn } x \Rightarrow (x^1 + 1^{23})^4) (3)^5)^6$$

$\hat{C}(1)$	$\emptyset$ (3)
$\hat{C}(2)$	$\emptyset$ (1)
$\hat{C}(3)$	$\emptyset$ (4)
$\hat{C}(4)$	$\{\text{fn } x \Rightarrow (x^1 + 1^{23})^4\}$
$\hat{C}(5)$	$\emptyset$ (3)
$\hat{C}(6)$	$\emptyset$ (4)
$\hat{\rho}(x)$	$\emptyset$ (3)

Exercise 1

$$[app] (\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2}$$

$$(\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) : (\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge \hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$$

Consider and add labels to the following expression
Guess an analysis result and verify it

$$e \triangleq ((\text{fn } x \Rightarrow (x^1 + 1^2)^3)^4 (3)^5)^6$$

$$(\hat{C}, \hat{\rho}) \models e \text{ iff } (\hat{C}, \hat{\rho}) \models (\text{fn } x \Rightarrow (x^1 + 1^2)^3)^4 \text{ iff } (\hat{C}, \hat{\rho}) \models \{\text{fn } x \Rightarrow (x^1 + 1^2)^3\} \subseteq \hat{C}(4)$$

$$\wedge (\hat{C}, \hat{\rho}) \models (3)^5 \text{ iff } \emptyset \subseteq \hat{C}(5)$$

$$\{\text{fn } x \Rightarrow (x^1 + 1^2)^3\} \subseteq \hat{C}(4) \Rightarrow (\hat{C}, \hat{\rho}) \models (x^1 + 1^2)^3 \wedge$$

$$\hat{C}(5) \subseteq \hat{\rho}(x) \wedge \hat{C}(3) \subseteq \hat{C}(6)$$

$$(\hat{C}, \hat{\rho}) \models (x^1 + 1^2)^3 \text{ iff } (\hat{C}, \hat{\rho}) \models x^1 \text{ iff } \hat{\rho}(x) \subseteq \hat{C}(1)$$

$$\wedge (\hat{C}, \hat{\rho}) \models 1^2 \text{ iff } \emptyset \subseteq \hat{C}(2)$$

Exercise 1

$$[app] (\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^\ell \text{ iff } (\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2}$$

$$(\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) : (\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge \hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$$

Consider and add labels to the following expression
Guess an analysis result and verify it

$$e \triangleq ((\text{fn } x \Rightarrow (x^1 + 1^2)^3)^4 (3)^5)^6$$

$$\{\text{fn } x \Rightarrow (x^1 + 1^2)^3\} \subseteq \hat{C}(4) \quad \checkmark$$

$$\emptyset \subseteq \hat{C}(5) \quad \checkmark$$

$$\hat{C}(5) \subseteq \hat{\rho}(x) \quad \checkmark$$

$$\hat{C}(3) \subseteq \hat{C}(6) \quad \checkmark$$

$$\hat{\rho}(x) \subseteq \hat{C}(1) \quad \checkmark$$

$$\emptyset \subseteq \hat{C}(2) \quad \checkmark$$

$\hat{C}(1)$	$\emptyset$ (3)
$\hat{C}(2)$	$\emptyset$ (1)
$\hat{C}(3)$	$\emptyset$ (4)
$\hat{C}(4)$	$\{\text{fn } x \Rightarrow (x^1 + 1^2)^3\}$
$\hat{C}(5)$	$\emptyset$ (3)
$\hat{C}(6)$	$\emptyset$ (4)
$\hat{\rho}(x)$	$\emptyset$ (3)



Exercise 2

Consider and add labels to the following expression: guess an analysis result and verify it

$$e \triangleq ((\text{fn } f \Rightarrow ((f^{1} 76^{2})^{3} + (f^{4} 77^{5})^{6})^{7})^{8} (\text{fn } a \Rightarrow a^{9})^{10})^{12}$$

Exercise 2

$$e \triangleq ((\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7)^8 (\text{fn } a \Rightarrow a^9)^{10})^{12}$$

$$\hat{C}(1) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{C}(2) = \emptyset \text{ (76)}$$

$$\hat{C}(3) = \emptyset \text{ (76)}$$

$$\hat{C}(4) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{C}(5) = \emptyset \text{ (77)}$$

$$\hat{C}(6) = \emptyset \text{ (77)}$$

$$\hat{C}(7) = \emptyset \text{ (76+77)}$$

$$\hat{C}(8) = \{\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^2\}$$

$$\hat{C}(9) = \emptyset \text{ (76, 77)}$$

$$\hat{C}(10) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{C}(11) = \emptyset$$

$$\hat{C}(12) = \{\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^2\}$$

$$\hat{\rho}(f) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{\rho}(a) = \emptyset \text{ (76, 77)}$$

Exercise 2

[app] $(\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^\ell$ iff $(\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2}$
 $(\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) :$
 $(\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge$
 $\hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$

$$e \triangleq ((\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7)^8 (\text{fn } a \Rightarrow a^9)^{10})^{12}$$

$$\begin{aligned} (\hat{C}, \hat{\rho}) \models e &\text{ iff } (\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7)^8 \text{ iff } \{\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7\} \in \hat{C}(8) \\ &\wedge (\hat{C}, \hat{\rho}) \models (\text{fn } a \Rightarrow a^9)^{10} \text{ iff } \{\text{fn } a \Rightarrow a^9\} \in \hat{C}(10) \wedge \\ &\wedge (\hat{C}, \hat{\rho}) \models ((f^1 76^2)^3 + (f^4 77^5)^6)^7 \text{ iff } (\hat{C}, \hat{\rho}) \models (f^1 76^2)^3 \wedge (\hat{C}, \hat{\rho}) \models (f^4 77^5)^6 \\ &\wedge \hat{C}(10) \subseteq \hat{\rho}(f) \wedge \hat{C}(8) \subseteq \hat{C}(12) \end{aligned}$$

$$\begin{aligned} (\hat{C}, \hat{\rho}) \models (f^1 76^2)^3 &\text{ iff } (\hat{C}, \hat{\rho}) \models f^1 \wedge (\hat{C}, \hat{\rho}) \models 76^2 \wedge (\hat{C}, \hat{\rho}) \models a^9 \wedge \hat{C}(2) \subseteq \hat{\rho}(a) \wedge \hat{C}(9) \subseteq \hat{C}(3) \\ (\hat{C}, \hat{\rho}) \models (f^4 77^5)^6 &\text{ iff } (\hat{C}, \hat{\rho}) \models f^4 \wedge (\hat{C}, \hat{\rho}) \models 77^5 \wedge (\hat{C}, \hat{\rho}) \models a^9 \wedge \hat{C}(5) \subseteq \hat{\rho}(a) \wedge \hat{C}(9) \subseteq \hat{C}(6) \\ (\hat{C}, \hat{\rho}) \models f^1 &\text{ iff } \hat{\rho}(f) \subseteq \hat{C}(1) \wedge (\hat{C}, \hat{\rho}) \models a^9 \text{ iff } \hat{\rho}(a) \subseteq \hat{C}(9) \\ (\hat{C}, \hat{\rho}) \models f^4 &\text{ iff } \hat{\rho}(f) \subseteq \hat{C}(4) \wedge \end{aligned}$$

Exercise 2

$((\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^7)^8 (\text{fn } a \Rightarrow a^9)^{10})^{12}$

$\{\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^2)\} \in \hat{C}(8)$

$\{\text{fn } a \Rightarrow a^9\} \in \hat{C}(10)$

$\hat{C}(10) \subseteq \hat{\rho}(f)$

$\hat{C}(8) \subseteq \hat{C}(12)$

$\hat{C}(2) \subseteq \hat{\rho}(a)$

$\hat{C}(9) \subseteq \hat{C}(3)$

$\hat{C}(9) \subseteq \hat{C}(6)$

$\hat{\rho}(f) \subseteq \hat{C}(1)$

$\hat{\rho}(f) \subseteq \hat{C}(4)$

$\hat{\rho}(a) \subseteq \hat{C}(9)$

$\hat{C}(1) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{C}(2) = \emptyset \text{ (76)}$

$\hat{C}(3) = \emptyset \text{ (76)}$

$\hat{C}(4) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{C}(5) = \emptyset \text{ (77)}$

$\hat{C}(6) = \emptyset \text{ (77)}$

$\hat{C}(7) = \emptyset \text{ (76+77)}$

$\hat{C}(8) = \{\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^2)\}$

$\hat{C}(9) = \emptyset \text{ (76, 77)}$

$\hat{C}(10) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{C}(11) = \emptyset$

$\hat{C}(12) = \{\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^2)\}$

$\hat{\rho}(f) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{\rho}(a) = \emptyset \text{ (76, 77)}$



Exercise 3

Consider and add labels to the following expression

Guess an analysis result and verify it

```
let f = fn x => x 1
  in let g = fn y => y + 2
    in let h = fn z => z + 3
      in (f g) + (f h)
```

$\text{id}_w \triangleq \text{fn } w \Rightarrow w^{\ell}$

```
e  $\triangleq$  (let f = (fn x => (x1 12)3)4
  in (let g = (fn y => (y5 + 26)7)8
    in (let h = (fn z => (z9 + 310)11)12
      in ((f13 g14)15 + (f16 h17)18)19)20)21)22
```

Complete the exercise

Theoretical properties

Semantic correctness

Goal: show that acceptable CFA results are semantically correct

The analysis information is a safe description of what will happen during the evaluation of the program, i.e., a **sound over-approximation** of the runtime behaviour

To reason on correctness, we define a **Small Step Structural Operational Semantics** for FUN, where:

- We use **intermediate expressions** *ie* to handle the binding of local variables
- We choose an approach based on **explicit environments** (matching the CFA abstraction) rather than substitutions

Subject reduction result

If the abstract ($\hat{\rho}$) and concrete environments (ρ) agree ($\rho \ R \ \hat{\rho}$):

$(\hat{C}, \hat{\rho}) \models ie$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie'$

to preserve the identity of
functions (and hence
abstract values) during
evaluation

$\rho \ R \ \hat{\rho}$ iff $dom(\rho) \subseteq dom(\hat{\rho}) \wedge \forall x \in dom(\rho) \ \forall t_x \ \forall \rho_x :$
 $(\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x \ R \ \hat{\rho})$

Configurations

A function definition will evaluate to a **closure** containing

- the syntax of the function definition, together with
- an environment mapping its free variables to their values

$v \in \mathbf{Val}$: values (evaluation results)

$\rho \in \mathbf{Env}$: environments (mapping free variables to their *values*)

$v ::= c \mid \text{close } t \text{ in } \rho$

$\rho ::= [] \mid \rho[x \mapsto v]$

Static binding:
functions evaluate to
closures capturing their
environment

Structural Operational Semantic

Terms and expressions

$ie \in \mathbf{IExp}$: intermediate expressions

$it \in \mathbf{ITerm}$: intermediate terms

$ie ::= it^\ell$

$t ::= c \mid x \mid \text{fn } x \Rightarrow e_0 \mid \text{fun } f \ x \Rightarrow e_0$
| $ie_1 \ ie_2$ | if ie_0 then e_1 else e_2 | let $x = ie_1$ in e_2 | ie_1 op ie_2
| bind ρ in ie // bind the environment (encoding of frames)
| close t in ρ // closure

Transitions: $\boxed{\rho \vdash ie \rightarrow ie'}$ one step of computation of the expression ie

Rule for variables

$$[var] \rho \vdash x^\ell \rightarrow v^\ell \text{ iff } x \in Dom(\rho) \text{ and } \rho(x) = v$$

The value of a variable is obtained from the environment

Example

$$\rho \vdash x^1 \rightarrow 7^1 \text{ if } x \in Dom(\rho) \text{ and } \rho(x) = 7$$

Rule for functions

$$[fn] \rho \vdash (fn\ x \Rightarrow e_0)^\ell \rightarrow (close\ (fn\ x \Rightarrow e_0)\ in\ \rho_0)^\ell$$

where $\rho_0 = \rho \mid FV(fn\ x \Rightarrow e_0)$

Static scoping:
freeze the environment
at definition time

This axiom constructs the appropriate closure

It restricts the environment ρ to the free variables of the abstraction

Note that the label ℓ is preserved in the closure

Example

$$[y \mapsto 7] \vdash (fn\ x \Rightarrow (x^1\ y^{23})^4) \rightarrow (close\ (fn\ x \Rightarrow (x^1\ y^{23}))\ in\ [y \mapsto 7])^4$$

$\rho_0 = \rho \mid \{y\} = [y \mapsto 7]$, since $FV(fn\ x \Rightarrow (x^1\ y^{23})) = \{y\}$

only free variables
are kept

Rule for let binding

$$[let_2] \ \rho \vdash (\text{let } x = v^{\ell_1} \text{ in } e_2)^{\ell} \rightarrow (\text{bind } \rho_0[x \mapsto v] \text{ in } e_2)^{\ell}$$

where $\rho_0 = \rho \upharpoonright FV(e_2)$

The expression e_2 is evaluated in the environment augmented with the new binding

Example

$$[g \mapsto (\text{fn } y \Rightarrow y^1)^2] \vdash (\text{let } f = (\text{fn } x \Rightarrow x^1)^2 \text{ in } (f^3 3^4)^5)$$
$$\rightarrow (\text{bind } \rho_0[f \mapsto (\text{fn } x \Rightarrow x^1)^2] \text{ in } (f^3 3^4)^5_2, \text{ where } \rho_0 = \rho \upharpoonright \{f\} = []$$

Rule for application

$$[app_{fn}] \rho \vdash ((\text{close}(\text{fn } x \Rightarrow e_1) \text{ in } \rho_1)^{\ell_1} v_2^{\ell_2})^{\ell} \rightarrow (\text{bind } \rho_1[x \mapsto v_2] \text{ in } e_1)^{\ell}$$

This axiom describes the binding of the actual parameter to the formal parameter

Example

$$[y \mapsto 5] \vdash ((\text{close}(\text{fn } x \Rightarrow (x^1 + y^2)^3)^4 \text{ in } [y \mapsto 7])^4 3^5)^6 \rightarrow (\text{bind } [y \mapsto 7, x \mapsto 3] \text{ in } (x^1 + y^2)^3)^6_{83}, \text{ where } y = 7 \text{ (not 5)}$$

Static scoping:
we use the definition
environment, not the call

Rules for binding

$$[bind_1] \frac{\rho \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (\text{bind } \rho_1 \text{ in } ie_1)^\ell \rightarrow (\text{bind } \rho_1 \text{ in } ie'_1)^\ell}$$

$$[bind_2] \rho \vdash (\text{bind } \rho_1 \text{ in } v_1^{\ell_1})^\ell \rightarrow v_1^\ell$$

- The bind construct is evaluated by repeatedly applying $[bind_1]$
- The final evaluation result is get by using rule $[bind_2]$

Example

$$[y \mapsto 5] \vdash (\text{bind } [y \mapsto 7, x \mapsto 3] \text{ in } x^1 + y^2)^6 \rightarrow * 10^6$$

Structural Operational Semantic I

$$\begin{array}{ll}
 [\text{var}] & \rho \vdash x^\ell \rightarrow v^\ell \text{ if } x \in \text{dom}(\rho) \text{ and } v = \rho(x) \\
 [\text{fn}] & \rho \vdash (\text{fn } x \Rightarrow e_0)^\ell \rightarrow (\text{close } (\text{fn } x \Rightarrow e_0) \text{ in } \rho_0)^\ell \\
 & \text{where } \rho_0 = \rho \mid FV(\text{fn } x \Rightarrow e_0) \\
 [\text{fun}] & \rho \vdash (\text{fun } f \ x \Rightarrow e_0)^\ell \rightarrow (\text{close } (\text{fun } f \ x \Rightarrow e_0) \text{ in } \rho_0)^\ell \\
 & \text{where } \rho_0 = \rho \mid FV(\text{fun } f \ x \Rightarrow e_0) \\
 [\text{app}_1] & \frac{\rho \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (ie_1 \ ie_2)^\ell \rightarrow (ie'_1 \ ie_2)^\ell} \\
 [\text{app}_2] & \frac{\rho \vdash ie_2 \rightarrow ie'_2}{\rho \vdash (v_1^{\ell_1} \ ie_2)^\ell \rightarrow (v_1^{\ell_1} \ ie'_2)^\ell} \\
 [\text{app}_{fn}] & \rho \vdash ((\text{close } (\text{fn } x \Rightarrow e_1) \text{ in } \rho_1)^{\ell_1} \ v_2^{\ell_2})^\ell \rightarrow \\
 & \quad (\text{bind } \rho_1[x \mapsto v_2] \text{ in } e_1)^\ell \\
 [\text{app}_{fun}] & \rho \vdash ((\text{close } (\text{fun } f \ x \Rightarrow e_1) \text{ in } \rho_1)^{\ell_1} \ v_2^{\ell_2})^\ell \rightarrow \\
 & \quad (\text{bind } \rho_2[x \mapsto v_2] \text{ in } e_1)^\ell \\
 & \quad \text{where } \rho_2 = \rho_1[f \mapsto \text{close } (\text{fun } f \ x \Rightarrow e_1) \text{ in } \rho_1] \\
 [\text{bind}_1] & \frac{\rho_1 \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (\text{bind } \rho_1 \text{ in } ie_1)^\ell \rightarrow (\text{bind } \rho_1 \text{ in } ie'_1)^\ell} \\
 [\text{bind}_2] & \rho \vdash (\text{bind } \rho_1 \text{ in } v_1^{\ell_1})^\ell \rightarrow v_1^\ell
 \end{array}$$

Structural Operational Semantic II

$$\begin{array}{l} [if_1] \quad \frac{\rho \vdash ie_0 \rightarrow ie'_0}{\rho \vdash (\text{if } ie_0 \text{ then } e_1 \text{ else } e_2)^\ell \rightarrow (\text{if } ie'_0 \text{ then } e_1 \text{ else } e_2)^\ell} \\ [if_2] \quad \rho \vdash (\text{if true}^{\ell_0} \text{ then } t_1^{\ell_1} \text{ else } t_2^{\ell_2})^\ell \rightarrow t_1^\ell \\ [if_3] \quad \rho \vdash (\text{if false}^{\ell_0} \text{ then } t_1^{\ell_1} \text{ else } t_2^{\ell_2})^\ell \rightarrow t_2^\ell \\ [let_1] \quad \frac{\rho \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (\text{let } x = ie_1 \text{ in } e_2)^\ell \rightarrow (\text{let } x = ie'_1 \text{ in } e_2)^\ell} \\ [let_2] \quad \rho \vdash (\text{let } x = v^{\ell_1} \text{ in } e_2)^\ell \rightarrow (\text{bind } \rho_0[x \mapsto v] \text{ in } e_2)^\ell \\ \quad \text{where } \rho_0 = \rho \mid FV(e_2) \\ [op_1] \quad \frac{\rho \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (ie_1 \text{ op } ie_2)^\ell \rightarrow (ie'_1 \text{ op } ie_2)^\ell} \\ [op_2] \quad \frac{\rho \vdash ie_2 \rightarrow ie'_2}{\rho \vdash (v_1^{\ell_1} \text{ op } ie_2)^\ell \rightarrow (v_1^{\ell_1} \text{ op } ie'_2)^\ell} \\ [op_3] \quad \rho \vdash (v_1^{\ell_1} \text{ op } v_2^{\ell_2})^\ell \rightarrow v^\ell \quad \text{if } v = v_1 \text{ op } v_2 \end{array}$$

Back to Example 1: evaluation

$((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5$

$[] \vdash ((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5$
→ $((\text{close}(\text{fn } x \Rightarrow x^1) \text{ in } [])^2 (\text{fn } y \Rightarrow y^3)^4)^5$
→ $((\text{close}(\text{fn } x \Rightarrow x^1) \text{ in } [])^2 (\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])^4)^5$
→ $(\text{bind } [x \mapsto (\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])] \text{ in } x^1)^5$
→ $(\text{bind } [x \mapsto (\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])] \text{ in } (\text{close } (\text{fn } y \Rightarrow y^3)^4 \text{ in } [])^1)^5$
→ $(\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])^5$

New CFA clauses

$$\begin{aligned} [bind] \quad (\hat{C}, \hat{\rho}) \models (\text{bind } \rho \text{ in } it_0^{\ell_0})^\ell &\text{ iff } (\hat{C}, \hat{\rho}) \models it_0^{\ell_0} \wedge \\ &\hat{C}(\ell_0) \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho} \\ [close] \quad (\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^\ell &\text{ iff } \{t_0\} \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho} \end{aligned}$$

$$\boxed{\begin{aligned} \rho R \hat{\rho} &\text{ iff } \text{dom}(\rho) \subseteq \text{dom}(\hat{\rho}) \wedge \forall x \in \text{dom}(\rho) \, \forall t_x \, \forall \rho_x : \\ &(\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x R \hat{\rho}) \end{aligned}}$$

For the other clauses just replace e with ie

Relation between environments

$$\rho \ R \ \hat{\rho} \text{ iff } \text{dom}(\rho) \subseteq \text{dom}(\hat{\rho}) \wedge \forall x \in \text{dom}(\rho) \ \forall t_x \ \forall \rho_x : \\ (\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x \ R \ \hat{\rho})$$

- The concrete domain must be included in the abstract one (is an over-approximation)
- For all variables $x \in \text{dom}(\rho)$, if $\rho(x) = \text{close } t_x \text{ in } \rho_x$ then
 - $t_x \in \hat{\rho}(x)$: the function is predicted
 - $\rho_x \ R \ \hat{\rho}$: the captured environment ρ_x is also correct
 - This second condition is recursive: ρ_x may contain closures, whose environments must also be checked

The abstract environment over-approximates every concrete closure, including all environments they recursively capture

Relation between values

The definition of R can be split into two mutually recursive definitions

$$v \ V (\hat{\rho}, \hat{v}) \text{ iff } \forall t \ \forall \rho : (v = \text{close } t \text{ in } \rho) \Rightarrow (t \in \hat{v} \wedge \rho \ R \hat{\rho})$$

$$\rho \ R \hat{\rho} \text{ iff } \text{dom}(\rho) \subseteq \text{dom}(\hat{\rho}) \wedge \forall x \in \text{dom}(\rho) : \rho(x) \ V (\hat{\rho}, \hat{\rho}(x))$$

The two definitions of R are equivalent. We focus on $v \ V (\hat{\rho}, \hat{v})$

- The values must agree as well
- For all t , and for all ρ in $(v = \text{close } t \text{ in } \rho)$
 - $t_x \in \hat{v}$: the value is predicted
 - $\rho_x \ R \hat{\rho}$: the captured environment ρ_x is also correct
 - This check is recursive: environments may contain closures

A concrete value is a closure, so we must approximate both the function and all the environments it captures

Subject reduction

Theorem

If $\rho \ R \ \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Idea: if there is a possible evaluation of the program s.t. the function at a call point evaluates to some abstraction, then this abstraction has to be in the set of possible abstractions computed by the analysis

All acceptable analysis results remain acceptable under evaluation

Proof idea

We assume that $\rho \ R \ \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ and prove $(\hat{C}, \hat{\rho}) \models ie'$ by **induction** on the structure of the **inference tree** for $\rho \vdash ie \rightarrow ie'$

For each rule, we show CFA is preserved

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Proof

We assume that $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ and prove $(\hat{C}, \hat{\rho}) \models ie'$ by **induction** on the structure of the **inference tree** for $\rho \vdash ie \rightarrow ie'$

The case **[var]**. Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash x^\ell \rightarrow v^\ell$ because $x \in \text{dom}(\rho) \wedge v = \rho(x)$

[Case $v = c$]: nothing to prove.

[Case $v = \text{close } t_0 \text{ in } \rho_0$].

- From $\rho R \hat{\rho}$ we get $v V (\hat{\rho}, \hat{\rho}(x))$ and hence $t_0 \in \hat{\rho}(x)$ and $\rho_0 R \hat{\rho}$.
- From $(\hat{C}, \hat{\rho}) \models ie$ we get $\hat{\rho}(x) \subseteq \hat{C}(\ell)$, and hence $t_0 \in \hat{C}(\ell)$.

Since $t_0 \in \hat{C}(\ell)$ and $\rho_0 R \hat{\rho}$ we have established $(\hat{C}, \hat{\rho}) \models ie'$.

[var] $(\hat{C}, \hat{\rho}) \models x^\ell$ iff $\hat{\rho}(x) \subseteq \hat{C}(\ell)$

$v V (\hat{\rho}, \hat{v})$ iff $\forall t \forall \rho : (v = \text{close } t \text{ in } \rho) \Rightarrow (t \in \hat{v} \wedge \rho R \hat{\rho})$

the value is already
predicted by the
analysis

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Proof

The case **[fn]**. Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (\text{fn } x \Rightarrow e_0)^\ell \rightarrow (\text{close } (\text{fn } x \Rightarrow e_0) \text{ in } \rho_0)^\ell$ where $\rho_0 = \rho \upharpoonright FV(\text{fn } x \Rightarrow e_0)$

- From $(\hat{C}, \hat{\rho}) \models ie$ we get $\{\text{fn } x \Rightarrow e_0\} \subseteq \hat{C}(\ell)$.
- From $\rho R \hat{\rho}$ we get we immediately get $\rho_0 R \hat{\rho}$ and then we can establish $(\hat{C}, \hat{\rho}) \models ie'$.

the abstraction is already recorded in $\hat{C}(\ell)$

[fn] $(\hat{C}, \hat{\rho}) \models (\text{fn } x \Rightarrow e_0)^\ell$ iff $\{\text{fn } x \Rightarrow e_0\} \subseteq \hat{C}(\ell)$

[close] $(\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^\ell$ iff $\{t_0\} \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Proof

The case **[app₁]**. Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (ie_1 ie_2)^\ell \rightarrow (ie'_1 ie_2)^\ell$ because $\rho \vdash ie_1 \rightarrow ie'_1$

- From the induction hypothesis applied to $(\hat{C}, \hat{\rho}) \models ie_1$, $\rho R \hat{\rho}$ and $\rho \vdash ie_1 \rightarrow ie'_1$, we get $(\hat{C}, \hat{\rho}) \models ie'$.

This suffices to establish $(\hat{C}, \hat{\rho}) \models (ie'_1 ie_2)^\ell$.

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Proof

The case **[app_{fn}]**.

Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (\text{close } (\text{fn } x \Rightarrow t_0^{\ell_0}) \text{ in } \rho_1)^{\ell_1} v_2^{\ell_2} \rightarrow (\text{bind } \rho_1[x \mapsto v_2] \text{ in } t_0^{\ell_0})^{\ell}$

- From $(\hat{C}, \hat{\rho}) \models ie$ we have $(\hat{C}, \hat{\rho}) \models (\text{close } (\text{fn } x \Rightarrow t_0^{\ell_0}) \text{ in } \rho_1)^{\ell_1}$, which yields

$\{\text{fn } x \Rightarrow t_0^{\ell_0}\} \subseteq \hat{C}(\ell_1)$ and $\rho R \hat{\rho}$.

- Further, we have $(\hat{C}, \hat{\rho}) \models v_2^{\ell_2}$

[close] $(\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^{\ell}$ iff $\{t_0\} \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

[bind] $(\hat{C}, \hat{\rho}) \models (\text{bind } \rho \text{ in } it_0^{\ell_0})^{\ell}$ iff $(\hat{C}, \hat{\rho}) \models it_0^{\ell_0} \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

[case $v_2 = c$]: immediate.

[case $v_2 = \text{close } t_2 \text{ in } \rho_2$]: it follows immediately from $(\hat{C}, \hat{\rho}) \models v_2^{\ell_2}$.

we simulate all possible function calls

Finally, the first universally quantified formula of the definition of $(\hat{C}, \hat{\rho}) \models ie$ gives

$(\hat{C}, \hat{\rho}) \models t_0^{\ell_0}$, $\hat{C}(\ell_2) \subseteq \hat{C}(\ell)$ and $(\rho_1[x \mapsto v_2]) R \hat{\rho}$ leading to establish $(\hat{C}, \hat{\rho}) \models ie'$.

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Fact

$(\hat{C}, \hat{\rho}) \models it^{\ell_1}$ and $\hat{C}(\ell_1) \subseteq \hat{C}(\ell_2)$ then $(\hat{C}, \hat{\rho}) \models it^{\ell_2}$

Proof

The case **[bind₂]**. Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (\text{bind } \rho_1 \text{ in } v_1^{\ell_1})^\ell \rightarrow v_1^\ell$

• From $(\hat{C}, \hat{\rho}) \models ie$ we have $(\hat{C}, \hat{\rho}) \models v_1^{\ell_1}$ and $\hat{C}(\ell_1) \subseteq \hat{C}(\ell)$

The desired $(\hat{C}, \hat{\rho}) \models v_1^\ell$ follows from **Fact**.

This completes the proof. ■

[bind] $(\hat{C}, \hat{\rho}) \models (\text{bind } \rho \text{ in } it_0^{\ell_0})^\ell$ iff $(\hat{C}, \hat{\rho}) \models it_0^{\ell_0} \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

CFA predictions are stable under evaluation

Back to Example 1: correctness of the solution

$$\begin{array}{c} ((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5 \\ \downarrow^* \\ (\text{Close}(\text{fn } y \Rightarrow y^3) \text{ in } [])^5 \end{array}$$

$$\begin{array}{c} (\hat{C}, \hat{\rho}) \models ((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5 \\ \downarrow \text{By Subject Reduction Theorem} \\ (\hat{C}, \hat{\rho}) \models (\text{Close}(\text{fn } y \Rightarrow y^3) \text{ in } [])^5 \end{array}$$

Exercise: Verify that the result (the closure) is acceptable

Back to Example 1: correctness of the solution (cont.)

the closure carries an empty environment

$$(\hat{C}_e, \hat{\rho}_e) \models (\text{Close}(\text{fn } y \Rightarrow y^3) \text{ in } [])^5$$

iff

	$(\hat{C}_e, \hat{\rho}_e)$	$\{(\text{fn } y \Rightarrow y^3)\} \subseteq \hat{C}(5) \wedge [] R \hat{\rho}$
1	$\{\text{fn } y \Rightarrow y^3\}$	✓
2	$\{\text{fn } x \Rightarrow x^1\}$	
3	$\emptyset$	
4	$\{\text{fn } y \Rightarrow y^3\}$	✓
5	$\{\text{fn } y \Rightarrow y^3\}$	
x	$\{\text{fn } y \Rightarrow y^3\}$	
y	$\emptyset$	

trivially true

The solution $(\hat{C}_e, \hat{\rho}_e)$ correctly predicts the result of evaluation

$$\rho R \hat{\rho} \text{ iff } \text{dom}(\rho) \subseteq \text{dom}(\hat{\rho}) \wedge \forall x \in \text{dom}(\rho) \forall t_x \forall \rho_x : (\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x R \hat{\rho})$$

$$[\text{close}] (\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^\ell \text{ iff } \{t_0\} \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$$